Cell

The TCGA Legacy

Much More Than a Catalog



Norman "Ned" Sharpless
NCI/NIH

The Cancer Genome Atlas (TCGA) represents a key milestone in the National Cancer Institute's mission to reduce the burden of cancer suffering. Before TCGA, there was a growing sense in cancer research that we were nearing a complete understanding of the basic biology of cancer. TCGA ended that hubris, identifying whole new pathways that were very understudied in cancer biology—for example, the role of the KEAP1-NRF2-CUL3 axis in squamous malignancies or the high frequency of SWI/SNF complex subunit mutations in many different types of cancer. I am struck by how an effort that was envisioned as a cataloging of somatic cancer mutations ended up teaching so much basic biology.

In retrospect, TCGA was also a critical step toward realizing the promise of precision oncology, providing a "source code" for these efforts to enhance cancer care. Knowing the list of driver mutations in the universe of cancer-causing genetic events allowed for a matching of clinical outcomes with specific mutations. In turn, therapy could be tailored based on those driver mutations, leading to more effective, less toxic cancer treatment. The FDA approval and Centers for Medicare and Medicaid Services coverage of next-generation-sequencing-based tests for patients with advanced cancer marks important additional progress of this effort. Access to these new sequencing technologies for diagnosis and therapy is great news for patients and would not have been possible without the groundwork laid by TCGA.

Big Data, Big Insights



Han Liang
The University of Texas MD Anderson Cancer Center

TCGA provides the most comprehensive molecular portrait of human cancers, and its impact is longstanding and multifold. First, using cutting-edge genome characterization technologies, TCGA essentially generates nearly complete catalogs of key molecular aberrations of various types in each cancer type, laying a critical foundation for more focused efforts of biomarker discovery and drug development. Second, TCGA characterizes patient diversity based on integrated molecular data, revolutionizing the ways of defining cancer types or subtypes (which are traditionally based on disease sites and pathology). Third, through TCGA, novel biological elements, such as lncRNAs and RNA editing, are emerging as players in cancer development. Fourth, because of parallel "omics" data over many large patient cohorts, TCGA enables the system-level, quantitative understanding of tumor biology by integrating information from DNA and RNA to protein and across cancer types.

TCGA also has tremendous impact beyond specific scientific findings. It highlights the importance of team science and data sharing in the era of precision medicine. Built by a carefully coordinated, highly collaborative research consortium, the rich, open-access resource (including data and related analytic/visualization tools) has empowered the whole cancer research community. Finally, TCGA has put bioinformatics and computational biology in the driver's seat of cancer research. As a bioinformatician and computational biologist who grew up with TCGA, I feel extremely lucky and proud of being part of this legacy.

Transformations in Technology and Analysis



Carolyn Hutter
NIH/NHGRI

When considering the legacy of TCGA, I always reflect on the transformations in genomic technology, methodology, and data analysis that occurred over the course of the project. As just one example, when the TCGA pilot phase started in 2006, the cost of a whole-genome sequence was approximately $14 million. The idea of comprehensive molecular characterization of cancer truly was an audacious goal. Today, in the era of the $1,000 genome, molecular characterization of cancer has become common in both clinical and research settings. The past decade has seen similar remarkable advances in many areas of genomics, including technology development, data-sharing policies, and data science.

Importantly, these innovations did not just occur concurrently with TCGA. Many of these advances were specifically developed or enhanced to meet the goals of the TCGA project. From initial work pioneering next-generation sequencing technologies through more recent applications of reproducible cloud-based analysis, TCGA investigators have helped shape the field through their groundbreaking contributions. The technological methods, analytical pipelines, and data standards stemming from TCGA contribute to a larger scientific framework that allows for transformational scientific advances, such as bridging gaps between discovery and clinical use—not just in cancer genomics, but in the broader application of genomics to multiple disease areas.

Collaboration: It Is in Our DNA



Jean-Claude Zenklusen
NCI/NHI

Large-team collaborative science has been a mainstay of the particle physics field since the Manhattan Project, probably due to the cost of the experimental instrumentation needed to collect the required data. With completion of the Human Genome Project in 2003, a vast uncharted expanse of biology became open to exploration. However, in those early times, the cost of sequencing made it impossible for large-scale projects to be single-lab endeavors. That might have been the initial impetus for researchers to band together at the launching of TCGA, which promised access to complex data of unprecedented magnitude and depth and the opportunity to analyze it in a multidisciplinary collaborative space. Although this team-science approach was thrusted upon the biomedical research community by necessity, it soon became clear that the quality of research accomplished by working collaboratively greatly outweighs the needs for individual recognition. In that way, TCGA has become the model for how to structure many of the large-scale projects that followed (e.g., the Epigenome Project or LINCS). Today, despite dramatic decreases in the cost of sequencing, the scientific community is even more enthusiastic about tackling large-scale genomic projects not as separate laboratories or institutions, but as confederations of scientists focused on a deep understanding of the data. It may have taken biomedical science five decades longer than physics, but appreciation for the value of implementing a true team-science approach to unravel the complexity of human disease is definitely here to stay.

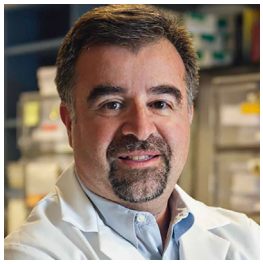
Maximizing Use of the Data



Hui Shen
Van Andel Research Institute

The deluge of data produced by TCGA came with computational and informatics challenges. One such challenge was ensuring easy and secure access for a broad base of users with varying levels of bioinformatics proficiency and access to computational resources and types of research questions. All TCGA data are hosted at the NCI-funded Genomic Data Commons (GDC), while further curated and summarized results are available through interactive user-friendly tools such as cBioPortal and Regulome Explorer. A centralized repository for large datasets presents challenges of data transfer and data security for local analyses, as well as disparities in available storage and computational power. A potential solution is to bring shared computational resources to the data rather than the other way around. In light of this, NCI launched the Cancer Genomic Cloud (CGC) pilot project in 2016, with TCGA data preloaded "in the cloud." Many of us previously involved in TCGA data production and analysis are now helping to enhance the utility of this GDC and CGC infrastructure. For instance, we have developed improved pipelines to reprocess and/or analyze TCGA data produced by our own team and are now moving toward making updated data packages available on GDC. We are also making our informatics tools readily transplantable for public use with cloud resources like the CGC. This is an ongoing process that will require participation by many. To quote Sir Winston Churchill, "Now this is not the end. It is not even the beginning of the end. But it is, perhaps, the end of the beginning."

Cancer Taxonomy Leaps Forward



Thomas J. Giordano
Michigan Medicine

Classification of human cancer represents one of the foundations of pathology and oncology. Tumors are evaluated and classified according to their histopathological features, which represents an enduring diagnostic platform that has served cancer patients and oncologists well for decades. This is not surprising, as the microscopic appearance of a tumor reflects the sum of its underlying genetic and epigenetic alterations. Yet we pathologists can do better as morphologically similar tumors can have distinct cancer genomes, an observation with critical significance for precision oncology.

TCGA began with three essential goals: to better understand the molecular underpinnings of cancer, to derive original molecular taxonomies, and to identify novel molecular therapeutic targets. While it has been clearly successful across all these domains, as a practicing pathologist, I see TCGA's impact on the taxonomy of human cancer as its crowning achievement. Nearly all TCGA projects identified previously unrecognized molecular subgroups, and moreover, genomic insights from some projects (e.g., thyroid cancer) have catalyzed significant taxonomic revisions. Collectively, by integrating tumor morphology with genomic features and molecular subtypes, TCGA has provided the knowledge to more accurately classify cancers. The resulting refined taxonomy of human cancer better reflects tumor biology. In the current era of precision oncology, these taxonomic advances will pay great dividends for cancer patients.